## 4.5 Universal joint

This is connected at both ends of the propeller shaft. Since the propeller shaft is connected by the universal joint, it transmits the engine's power at different angles.

## 4.6 Slip joint

It is connected with the propeller shaft after the universal joint. It moves smoothly according to the movement of rear axle. Figure 4.6 shows the view of a slip joint used in automobiles.

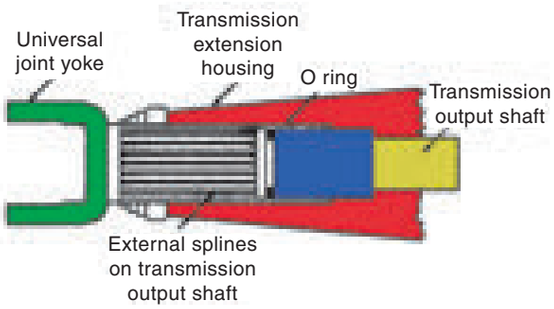



**Figure 4.6** Slip joint used in automobile

## 4.7 Functions of universal joint

- It transmits the power from the gear box shaft to the propeller shaft through inclined direction.
- It helps to transmit the power without any loss when the vehicle is going on ups and downs of the roads.
- It withstands the vibrations occurring on the roads.
- It helps in varying the length of the propeller shaft.

## 4.8 Types of universal joint

1. Hooke's type
2. Pot type
3. Rubber coupling type
4. Ball type

### 4.8.1 Hooke's type

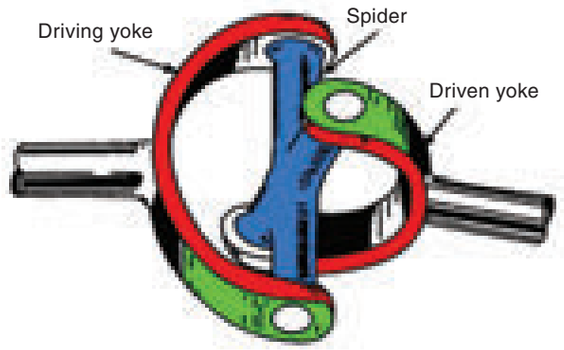



**Figure 4.8.1** Hooke type universal joint

In this type of joint the spider is fixed in between the two yokes. In the spider, needle like bearings are present. These bearings are set at the bottom of the yoke on the slots made on the yoke for placing the bearings. Hence the bearings cannot come out of the yoke. One yoke is connected with the gear box and another one is connected with the propeller shaft. The spider rotates along with the driving yoke which is connected to the gear box. Thereby the drive yoke which is connected to the other side of the spider starts to rotate and transmits the power from gear box to propeller shaft smoothly. Figure 4.7 shows the hooke's join.

#### Advantages

1. It has high mechanical advantage.
2. It is simple in construction.
3. It transmits the power more effectively.
4. It works according to the weight of the vehicle.
5. It has long life.



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